

Haein Jung, Ph.D.

Employment

- Aug. 2026 – Present  **Postdoctoral Researcher**, Brown University, CRUNCH Group.
Research focus: Boiling systems using AI agents, data-driven generative model, CFD, and continuum modeling.
- Sep. 2025 – Jul. 2026  **Postdoctoral Researcher**, Gwangju Institute of Science and Technology (GIST).
Research on laser thermal-fluid system design and high-performance energy systems.

Education

- Sep. 2019 – Aug. 2025  **Ph.D. in Mechanical and Robotics Engineering**, Gwangju Institute of Science and Technology (GIST), South Korea.
Dissertation: *Hydrodynamic Instabilities and Heat Transfer Behavior of Two-Phase Flow Boiling in Rectangular Microchannels and Concentric Annuli*.
- Sep. 2017 – Aug. 2019  **M.S. in Aerospace and Mechanical Engineering**, Korea Aerospace University, South Korea.
Thesis: *Chaotic advection in a groove-embedded partitioned pipe mixer: application to crossflow filtration*.
- Mar. 2010 – Aug. 2017  **B.S. in Mechanical Engineering**, Korea Aerospace University, South Korea.

Research Experience

- 2026–Present  **Physics Intelligence for Boiling**
Brown University, CRUNCH Group. Research on physics discovery AI agents, diffusion model for boiling system future forecasting, GPU-accelerated two-phase simulations, phase-field modeling, including research within the ONR METHODS MURI Thrust 3.
- 2024–2026  **30 kW Airborne Liquid-Immersion Solid-State Laser Thermal Management**
Agency for Defense Development (ADD), Korea. Experimental and computational analysis of heat transfer, pressure drop, thermal stress, and optical performance.
- 2019–2024  **540 kW Ultra-Fast EV Charging Cable Thermal Management**
Korea Electric Power Corporation (KEPCO), Korea. Experimental investigation of subcooled boiling heat transfer, pressure drop, CHF, and flow instability.
- 2017–2019  **Crossflow Filtration Fouling Reduction**
Computational investigation of chaotic mixing, pressure drop, and fouling mitigation.

Research Publications

Journal Articles

- 1 **H. Jung** and S.-H. Lee, “Experimental investigation of subcooled density-wave oscillations in concentric annulus,” *International Journal of Heat and Mass Transfer*, vol. 268, p. 128 967, 2026.
- 2 H. Noh, **H. Jung**, M. Kwon, and S.-H. Lee, “Comparison of subcooled flow boiling in concentric annuli with bare and finned surfaces,” *Applied Thermal Engineering*, vol. 258, p. 124 722, 2025.
- 3 **H. Jung** and S.-H. Lee, “Experimental investigation into thermal characteristics of subcooled flow boiling in horizontal concentric annuli for cooling ultra-fast electric vehicle charging cables,” *Case Studies in Thermal Engineering*, vol. 62, p. 105 122, 2024.
- 4 **H. Jung**, H. Noh, and S.-H. Lee, “Experimental and photographic investigation into horizontal subcooled flow boiling in concentric annuli for cooling system of ultra-fast electric vehicle charging cables,” *International Communications in Heat and Mass Transfer*, vol. 159, p. 108 059, 2024.
- 5 **H. Jung** and S.-H. Lee, “Mechanistic model to predict oscillating frequency of flow boiling in large length-to-diameter ratio micro-channel heat sinks,” *International Journal of Heat and Mass Transfer*, vol. 215, p. 124 473, 2023.
- 6 **H. Jung**, J.-M. Park, S.-Y. Jung, T.-G. Kang, and K.-H. Ahn, “Flow and mixing characteristics of a groove-embedded partitioned pipe mixer,” *Korea-Australia Rheology Journal*, vol. 32, pp. 319–329, 2020.
- 7 S.-Y. Jung, **H. Jung**, T.-G. Kang, and K.-H. Ahn, “Fouling mitigation in crossflow filtration using chaotic advection: A numerical study,” *AIChE Journal*, vol. 66, no. 1, e16792, 2020.
- 8 **H. Jung**, S.-Y. Jung, T.-G. Kang, and K.-H. Ahn, “Numerical study on the mixing in a barrier-embedded partitioned pipe mixer (bppm) for non-creeping flow conditions,” *Korea-Australia Rheology Journal*, vol. 30, pp. 227–238, 2018.

Skills

Experimental	Single-phase and two-phase flow test module and loop design, including 3D CAD (Autodesk Inventor, Creo), assembly, high-speed and IR imaging, electrical power-control box design, and data acquisition.
Theoretical	Continuum and mechanistic modeling of two-phase flow using conservation equations and closure relations, including 1D, 2D, and 3D steady-state and transient models.
Computational	CFD and high-fidelity simulation (ANSYS Fluent, Nektar phase-field solver), mesh generation (Gmsh, ANSYS Meshing), geometry and computational-domain generation (ANSYS SpaceClaim, DesignModeler), post-processing (CFD-Post, ParaView, Tecplot), and Linux HPC server management.
Data-Driven	Physics-informed neural operators, variational autoencoders, and generative diffusion models.
Intelligence-Driven	LLM-based agentic scientific analysis using commercial APIs, open-source foundation models, and CrewAI.

Additional Experience

Aug. 2011 – May 2013  **Military Service**, Republic of Korea Army.